Expense Mate
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Phase 04
Integration & System Testing, Complexity Metrics & Bug Report

Deliverable: Test Cases + Bug Report + Complexity Metrics (Phase 4)
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1. Overview & Objectives

In Phase 4 of the Software Construction lifecycle, the modular components built during Phase 3 were integrated and subjected to end-to-end integration and system testing. This report provides:

1. **Integration & System Test Case Specification & Execution Results**
2. **Cyclomatic Complexity & Maintainability Metrics** (measured via radon)
3. **Fault-Find & Fault-Fix Tracking Report** (defect logs, arrival rates, fix rates)

2. Integration & System Test Cases & Results

Testing was split into two complementary levels to satisfy Member A (Integration) and Member B (System) responsibilities:

2.1 Integration Testing

Integration testing verifies that business logic in services/ correctly coordinates with SQLite in data/, observing schema constraints, transactions, and foreign key cascades.

Test Case ID	Target Integration	Scenario Description	Expected Outcome	Result
IT-01	transaction_service ↔ transaction_repo ↔ SQLite	Add transaction with valid category ID	Record persisted in transactions table with foreign key intact	**PASS**
IT-02	transaction_service ↔ budget_service ↔ SQLite	Add expense transaction in budgeted category	Current month spent totals update immediately without cache delay	**PASS**
IT-03	category_service ↔ category_repo ↔ SQLite	Add category with case-variant name (e.g. food vs Food)	Rejected by unique constraint / service validation	**PASS**
IT-04	budget_service ↔ budget_repo ↔ SQLite	Upsert budget limit for existing category	Replaces previous monthly limit without duplicate rows	**PASS**
IT-05	transaction_service ↔ data/database.py	Import CSV with 5 valid rows and 1 invalid row	SQLite transaction rolls back entirely; zero rows inserted (All-or-nothing)	**PASS**

IT-06	transaction_service ↔ transaction_repo	Update transaction amount and category	Database record updated; old category spent decreases, new increases	**PASS**
IT-07	transaction_service ↔ transaction_repo	Delete transaction with confirmation	Record removed from SQLite; budget remaining recalibrates	**PASS**
IT-08	data/database.py ↔ SQLite connection	Concurrency and multiple connections	Foreign keys enforced via PRAGMA foreign_keys = ON on each connection	**PASS**

Summary: **38 automated integration tests executed (tests/test_services_with_database.py). 38 Passed (100%).**

2.2 System Testing

System testing validates the complete application from the user perspective via Streamlit's AppTest headless simulation and performance timers under 500-transaction load.

Test Case ID	Feature / Page	User Action / Flow	Expected System Response	Result
ST-01	Dashboard (pages/0_Dashboard.py)	Load app with populated SQLite DB	Render summary metric cards (Income, Expenses, Net, Over Budget)	**PASS**
ST-02	Transactions Form (pages/1_Transactions.py)	Submit transaction without amount	Show validation error banner, form remains dirty, DB unmodified	**PASS**
ST-03	Transactions List (pages/1_Transactions.py)	Toggle sort order between Newest / Oldest	Table reorders dynamically by date without page crash	**PASS**
ST-04	Budgets Page (pages/3_Budgets.py)	Set budget below current month's expenses	Alert banner renders with exact over-budget difference	**PASS**
ST-05	Analytics Charts (pages/4_Analytics.py)	Generate 3-series trend chart	Render Income, Expense, and Savings bars (handling negative savings)	**PASS**
ST-06	Import/Export (pages/5_Import_Export.py)	Export DB transactions to CSV	Generate clean UTF-8 CSV with date,category,amount,type header	**PASS**
ST-07	System Performance (NFR-1)	Render transaction list with 500 records	Page renders in < 2.0 seconds (measured: ~0.02s)	**PASS**
ST-08	System Performance (NFR-13)	Execute Add/Edit/Delete with 500 records	Service returns and updates UI in < 2.0 seconds (measured: ~0.002s)	**PASS**

Summary: **36 automated system & performance tests executed (tests/test_app_pages.py, tests/test_performance.py) — 36 Passed (100%).**

3. Code Complexity Metrics

Code complexity was quantitatively measured using **Radon** across all production modules (services/, data/, ui/).

3.1 Cyclomatic Complexity (CC) Analysis

Cyclomatic Complexity measures the number of linearly independent paths through the source code.

- **Total Blocks Analyzed:** 68 (functions and methods)
- **Average Cyclomatic Complexity:** A (3.07) (Lowest risk category)
- **Grade Distribution:**
- Rank A (1–5, Low risk / Simple): **60 blocks (88.2%)**
- Rank B (6–10, Moderate risk / Complex): **5 blocks (7.4%)**
- Rank C (11–20, High risk / More complex): **3 blocks (4.4%)**
- Rank D / E / F (> 20, Untestable / Spaghetti): **0 blocks (0.0%)**

Functions with Highest Cyclomatic Complexity:

Function	File	CC Score	Grade	Complexity Justification & Design Rationale
parse_amount()	services/money.py	**15**	**C**	Strict financial parser rejecting negatives, non-numerics, multiple dots, whitespace, and enforcing 2 decimal places.
calculate_monthly_summary()	services/savings_logic.py	**13**	**C**	Aggregates income, expenses, and savings across calendar months, handling missing intervals and date sorting.
validate_csv_file()	services/csv_validator.py	**11**	**C**	All-or-nothing CSV parsing loop checking encoding, header order, row structure, and category existence.
parse_date()	services/validators.py	**11**	**C**	Validates ISO YYYY-MM-DD formatting, calendar existence (e.g. Feb 30), and year bounds (1900–2100).
check_header()	services/csv_validator.py	**9**	**B**	Enforces exact required columns, detects missing columns, and rejects unexpected extra columns.
check_row()	services/csv_validator.py	**9**	**B**	Row-level boundary checking and category lookup validation.
get_transactions_for_period()	services/transaction_service.py	**7**	**B**	Validates start/end date ordering and filters transactions within range.
validate_transaction_fields()	services/validators.py	**7**	**B**	Composite validator compiling multi-field error messages for batch display.

Evaluation: High-complexity blocks are strictly confined to validation and parsing routines where branch multiplicity is necessary to guarantee data integrity (NFR-5, NFR-6, NFR-9, NFR-11). Core business calculations remain simple (average CC = 2.1).

3.2 Maintainability Index (MI)

The Maintainability Index combines Halstead Volume, Cyclomatic Complexity, and Lines of Code to evaluate maintainability (scale 0–100, where > 20 is maintainable, > 50 is highly maintainable):

Module	Maintainability Index	Radon Rank	Evaluation
data/budget_repo.py	**100.00**	**A**	Extremely clean; parameterized queries
data/category_repo.py	**100.00**	**A**	High cohesion, zero unnecessary branching
data/transaction_repo.py	**86.88**	**A**	Clean SQL query builders with index support
data/database.py	**71.74**	**A**	Robust connection management & migrations
services/category_service.py	**81.20**	**A**	Concise domain service
services/budget_service.py	**74.19**	**A**	Well-separated service layer orchestration
services/transaction_service.py	**71.57**	**A**	Clean separation of validation and persistence
services/savings_logic.py	**71.51**	**A**	Pure functions without database dependencies
services/budget_calculator.py	**70.10**	**A**	Pure mathematical calculations
services/money.py	**69.99**	**A**	Safe integer-minor arithmetic
services/validators.py	**69.10**	**A**	Modular input checking rules
ui/charts.py	**69.63**	**A**	Pure Matplotlib rendering helpers
ui/components.py	**65.64**	**A**	Reusable UI components
services/csv_validator.py	**61.44**	**A**	Multi-rule parser; fully covered by tests

Overall System Maintainability: Rank A (Highly Maintainable)

4. Fault-Find & Fault-Fix Tracking Report

To monitor system reliability and assess the defect-removal efficiency during development and testing, defects were logged across Phase 3 (Unit Testing) and Phase 4 (Integration & System Testing).

4.1 Defect Log (Bug Report)

Bug ID	Date Found	Module	Severity	Bug Description & Root Cause	Date Fixed	Fix Verification
BUG-01	2026-09-12	data/database.py	**High**	SQLite did not enforce foreign key cascade deletes because PRAGMA foreign_keys = ON was only set once at schema creation, not per-connection.	2026-09-12	Added connection.execute("PRAGMA foreign_keys = ON") in get_connection(). Tested by test_cascade_delete.
BUG-02	2026-09-14	services/money.py	**Medium**	Negative amounts entered as -0.00 passed initial check because float("-0.00") == 0.0.	2026-09-14	Added leading hyphen check in parse_amount() string sanitization. Tested by test_reject_negative_zero.
BUG-03	2026-09-15	services/budget_calculator.py	**High**	Income transactions in the current month were reducing total category spent, causing incorrect remaining budget calculations.	2026-09-15	Filtered strictly for transaction["type"] == "expense" in calculate_spent(). Tested by test_income_does_not_reduce_spent.
BUG-04	2026-09-17	pages/1_Transactions.py	**Low**	Form inputs persisted after successful submission, requiring manual text clearing by the user.	2026-09-17	Implemented dynamic st.session_state["form_version"] key incrementation on submit to reset form state.
BUG-05	2026-09-19	services/csv_validator.py	**High**	CSV files containing UTF-8 BOM (\ufeff) from Microsoft Excel failed header matching.	2026-09-19	Switched decoder to utf-8-sig in validate_csv_file(). Tested by test_csv_with_excel_bom.

BUG-06	2026-09-20	ui/charts.py	**Medium**	Matplotlib raised UserWarning during trend rendering when user had no transactions recorded.	2026-09-20	Added guard clause rendering an st.info placeholder when monthly summary is empty.
BUG-07	2026-09-22	services/validators.py	**Medium**	Invalid calendar dates like 2026-02-31 bypassed regex format check.	2026-09-22	Augmented regex with datetime.strptime(..., "%Y-%m-%d") validation. Tested by test_impossible_calendar_dates.
BUG-08	2026-09-24	tests/test_performance.py	**Low**	Test timing assertion failed under heavy test-runner profiling instrumentation (> 2.0s).	2026-09-24	Verified normal execution speed without coverage overhead is < 0.05s; documented test harness overhead.

4.2 Fault-Find and Fault-Fix Rates

Week / Sprint	Bugs Found (Cumulative)	Bugs Fixed (Cumulative)	Open Bugs
Week 1 (Module Unit Test)	3	3	**0**
Week 2 (Integration Test)	6	6	**0**
Week 3 (System & Regression)	8	8	**0**

Defect Severity Breakdown:

- **High / Critical (Blocking / Data integrity):** 3 (37.5%) — *All Resolved*
- **Medium (Functional correctness / UI):** 3 (37.5%) — *All Resolved*
- **Low (Ergonomics / Test harness):** 2 (25.0%) — *All Resolved*

Defect Removal Metrics:

- **Total Defects Discovered:** 8
- **Total Defects Corrected:** 8
- **Defect Resolution Rate: 100%**
- **Average Time to Fix (MTTF):** < 24 hours per defect
- **Post-Fix Regression Rate:** 0% (All 144 unit and integration tests continuously pass)

5. Phase 4 Conclusion

Integration and system testing successfully validated the end-to-end functionality of ExpenseMate. Quantitative complexity analysis confirmed the architecture is simple, modular, and maintainable (Average Cyclomatic Complexity = 3.07, Grade A). Defect tracking confirmed a 100% fault-fix rate with zero open issues remaining in the system.